19.3. FISSION PROBABILITY

When uranium or plutonium is irradiated by neutrons, in addition to neutron capture followed by fission (n,f), several different processes occur: scattering, (n,γ), (n,2n) reactions, etc. All these reactions are of importance for the reactor designer as well as for the chemists who have to manufacture new fuel pellets and take care of the spent fuel elements. The probability for the various reactions depends on the neutron energy. As is seen in Figure 19.3, three regions are clearly distinguishable:

(1) For thermal neutrons with average kinetic energies ($\bar{E}_n$) $\leq$1 eV, fission of ^{235}U and ^{239}Pu dominates over neutron capture (i.e. $\sigma_f > \sigma_{n,\gamma}$). Though $\sigma_{n,\gamma}$ for ^{238}U is small considerable capture occurs because of the large fraction of ^{238}U normally present.

(2) For epithermal neutrons ($1 \leq \bar{E}_n \leq 10^5$ eV) large radiative capture and fission resonances occur. In this region heavier isotopes are formed by (n,γ) reactions: ^{235}U(n,γ)^{236}U; ^{238}U(n,γ)^{239}U; ^{239}Pu(n,γ)^{240}Pu; ^{240}Pu(n,γ)^{241}Pu; etc (see Fig. 19.5).

(3) For fast neutrons ($\bar{E}_n \geq 0.1$ MeV) the cross sections are relatively small, $\leq$1 b. Fission dominates over radiative capture. Of particular importance is that ^{238}U becomes fissionable at a neutron energy of ~0.6 MeV; its fission cross section increases with neutron energy above the threshold to a constant value of ~0.5 b at $\geq$2 MeV.

It is obvious that the neutron energy spectrum of a reactor plays an essential role. Figure 19.4 shows the prompt (unmoderated) fission neutron spectrum with $\bar{E}_n$ ~2 MeV. In a nuclear explosive device almost all fission is caused by fast neutrons.

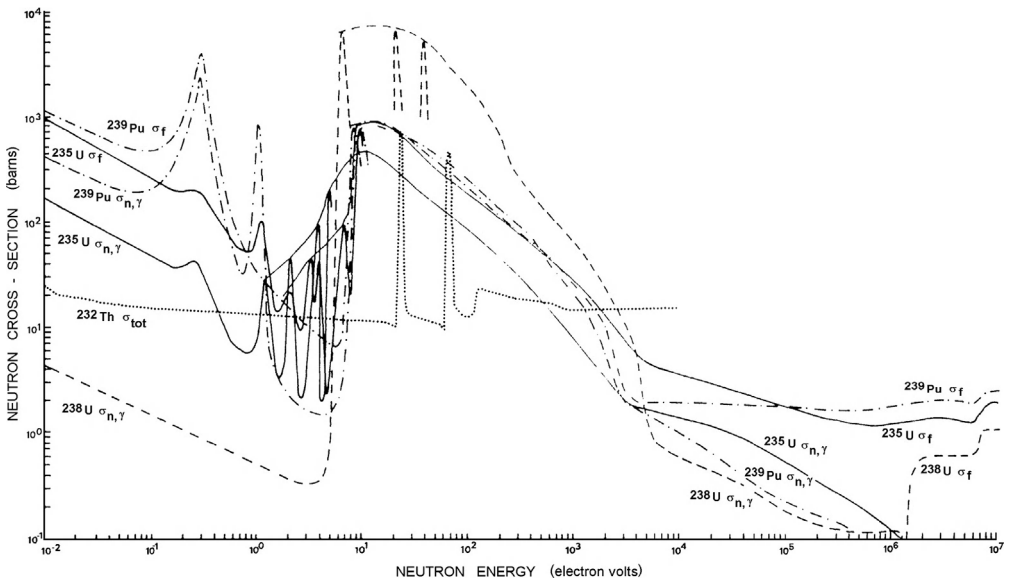


Figure 19.3 Cross sections for n-capture ($\sigma_{n,\gamma}$), fission (σ_f), and total (σ_{tot}) as a function of neutron energy.

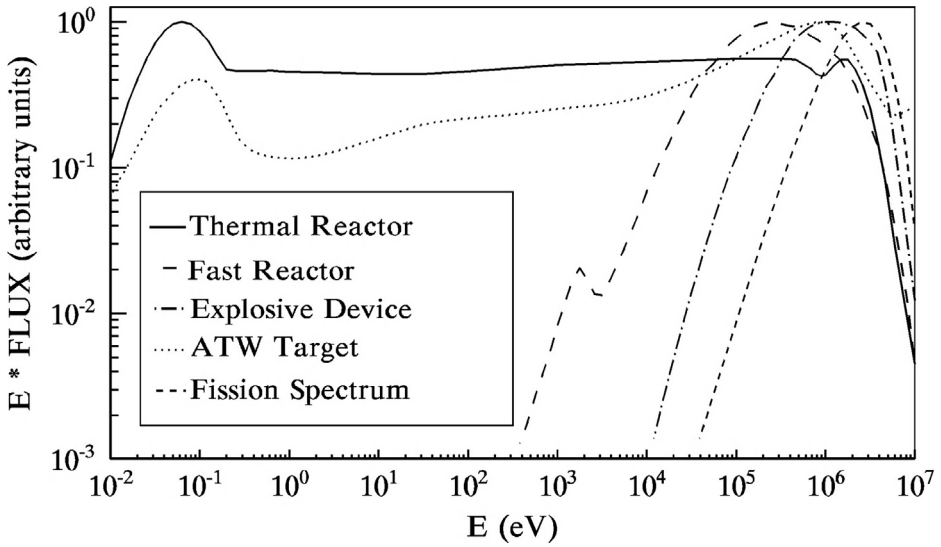


Figure 19.4 Neutron spectra of some chain reacting systems and a proposed accelerator driven system. The abscissa shows neutron flux times energy on an arbitrary scale.

Nuclear reactors can be designed so that fission mainly occurs with fast neutrons or mainly with slow neutrons (by slowing the neutrons to thermal energies before they encounter fuel). This leads to two different reactor concepts — the *fast reactor* and the *thermal reactor*. The approximate neutron spectra for both reactor types are shown in Figure 19.4. Because thermal reactors are more important at present, we discuss this type of reactors first.

In an ideal thermal reactor we may assume that most of the neutrons are in thermal equilibrium with the moderator atoms, though in practice the neutron spectrum in power reactors is much more energetic due to incomplete moderation. From $\bar{E}_n = \mathbf{k}T$ it follows that at about 300°C (a typical moderator temperature in a LWR) $\bar{E}_n$ is 0.05 eV; cf. Fig. 19.4. However, "thermal neutron energy" (E_{th}) cross sections are standardized to mono-energetic neutrons with a velocity of 2200 m s^{-1}, which corresponds to E_n 0.025 eV according to (3.32). Cross sections at this energy are given in Table 19.2. For most (but not all) nuclides the cross section at neutron energies below the resonance region may be estimated by the $1/v$ law (§11.4). Because thermal reactors have neutrons of more than one energy, *effective cross sections*[4] must be used for calculating reactor product yields. Such cross sections are usually calculated as an average over the core and vary with time as well as from reactor to reactor. Typical LWR effective cross sections are given in Table 19.2 and Figure 19.5 for thin targets. Self screening (depression of the n-flux due to a

[4] The effective cross-section is given by $\sigma_{eff} = \int\sigma(E)\ \varphi(E)\ dE\ /\int\varphi(E)\ dE$, where σ is cross-section, φ flux and E neutron energy.

Table 19.2 Nuclear data for some fissile and fertile nuclides (see also Figs 19.3, 19.5 and 20.5)

	^{232}Th	^{233}U	^{235}U	^{238}U	Nat. U	^{239}Pu	^{240}Pu	^{241}Pu
Radioactive decay	α,sf	α	α	α,sf	–	α,sf	α,sf	α,sf
Half-life (years)	1.405×10^{10}	1.592×10^{5}	7.038×10^{8}	4.466×10^{9}	–	2.411×10^{4}	6.563×10^{3}	14.35
Specific rad. act. (GBq/kg)	4.06×10^{-3}	356	0.0800	0.01244	–	2296	8398	3.83×10^{6}
Specific decay heat (W/kg)	0.003	275	0.056	0.008	–	1876	6950	3250
Thermal neutrons (0.025 eV)								
n,γ capture (σ_γ barns)	7.40	47.7	98.6	2.70	3.39	268.8	289.5	368
fission (σ_f barns)	39 μb	531.1	582.2	<0.5 mb	4.19	742.5	0.030	1009
neutron yield (ν)		2.492	2.418		2.418	2.871	2.90 ?	2.927
fission factor (η)		2.287	2.068		1.34	2.108	0.0003 ?	2.145
Fast neutrons (∼0.25; ∼1.0 MeV)								
n,γ-capture (σ_γ barns)	0.18; 0.12	0.22; 0.056	0.28; 0.11	0.15; 0.14	0.15; 0.14	0.17; 0.04	0.15; 0.18	2.0; 2.0
fission (σ_f barns)	; 0.067	2.20; 1.90	1.28; 1.19	; 0.018	0.01; 0.03	1.53; 1.73	0.10; 1.06	1.78; 1.56
neutron yield (ν)	; 2.1	2.503; 2.595	2.426; 2.522	; 2.50	2.426; 2.50	2.886; 3.001	2.80; 2.95	3.01; 3.12
fission factor (η)	; 0.75	2.28; 2.52	1.99; 2.31	; 0.28	0.15; 0.44	2.60; 2.93	1.13; 2.52	1.4; 1.4
Reactor spectrum neutrons (∼PWR; LMFBR[†])								
n,γ-capture (σ_γ barns)	4; 0.30	28; 0.23	107.5; 0.584	1.5; 0.271		559.5; 0.547	2616.8; 0.52	3428.7; 0.568
fission (σ_f barns)	; 0.01	298; 2.52	448.1; 1.951	; 0.040		1048.7; 1.81	; 0.371	1138.6; 2.64
neutron yield (ν)	; 2.4	2.49; 2.6	2.43; 2.445	; 2.900		2.87; 2.922	; 2.976	3.06; 2.968
fission factor (η)	; 0.08	2.27; 2.4	1.96; 1.882	; 0.37		1.86; 2.24	; 1.235	2.22; 2.443

[†]Creys-Malville central core.

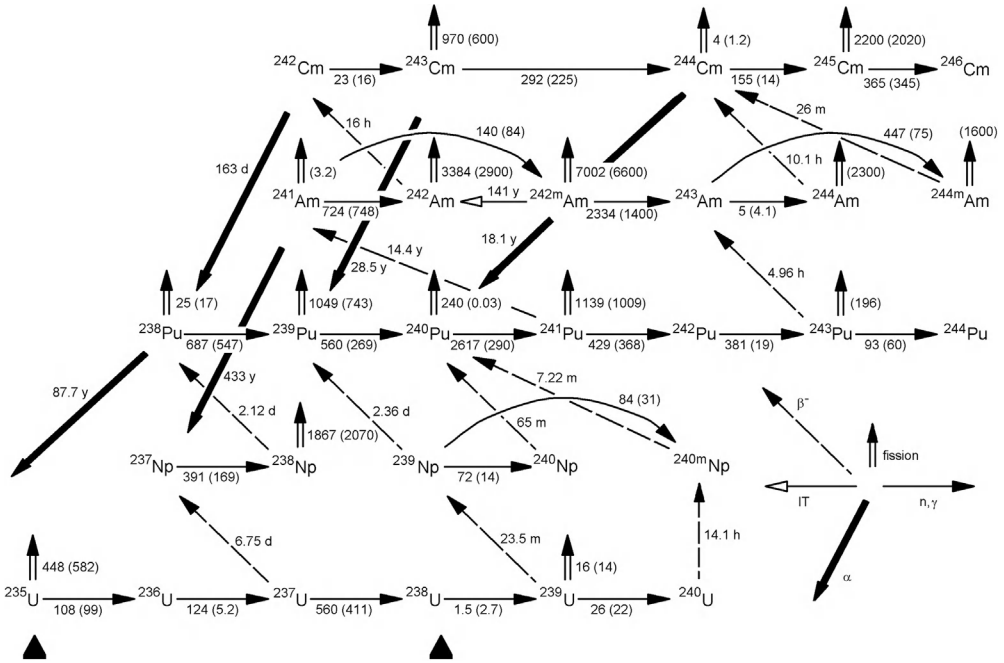


Figure 19.5 Nuclear reactions in irradiation of uranium. Figures along the arrows are half lives or effective cross-sections (b) for a typical power LWR, with thermal (0.025 eV) data within parentheses.

high Σ_a) reduces the effective cross sections below those given in Table 19.2 and Figure 19.5 for isotopes with high concentration in the fuel pins.

19.4. THE FISSION FACTOR

In thermal reactors ^{235}U is consumed mainly by fission and radiative capture

$$^{235}\mathrm{U} + \mathrm{n_{th}} \begin{cases} \xrightarrow[582\ \mathrm{b}]{\sigma_{n,f}} & 2\ \mathrm{FP} + \nu\mathrm{n} \\ \xrightarrow[99\ \mathrm{b}]{\sigma_{n,\gamma}} & ^{236}\mathrm{U} \\ & \downarrow \alpha \end{cases} \tag{19.4}$$

^{236}U decays through α-emission. Due to its long half-life (2.3×10^7 y) it accumulates in the reactor. ^{236}U may capture a neutron, forming ^{237}U, which over a few days time decays to ^{237}Np. The *neutron yield per fission*, ν, is an average value which depends on the neutron energy; it is 2.42 for thermal fission of ^{235}U (see Table 19.2).